Introduction to Programming

Lecture 7

VU
Virtual University

while loop

```
while (condition)
 {
   statements;     :
 }
statements;
```

While loop executes zero or more times. What if we want the loop to execute at least one time?

do-while

Do while loop execute on or more times

Syntax of do-while loop

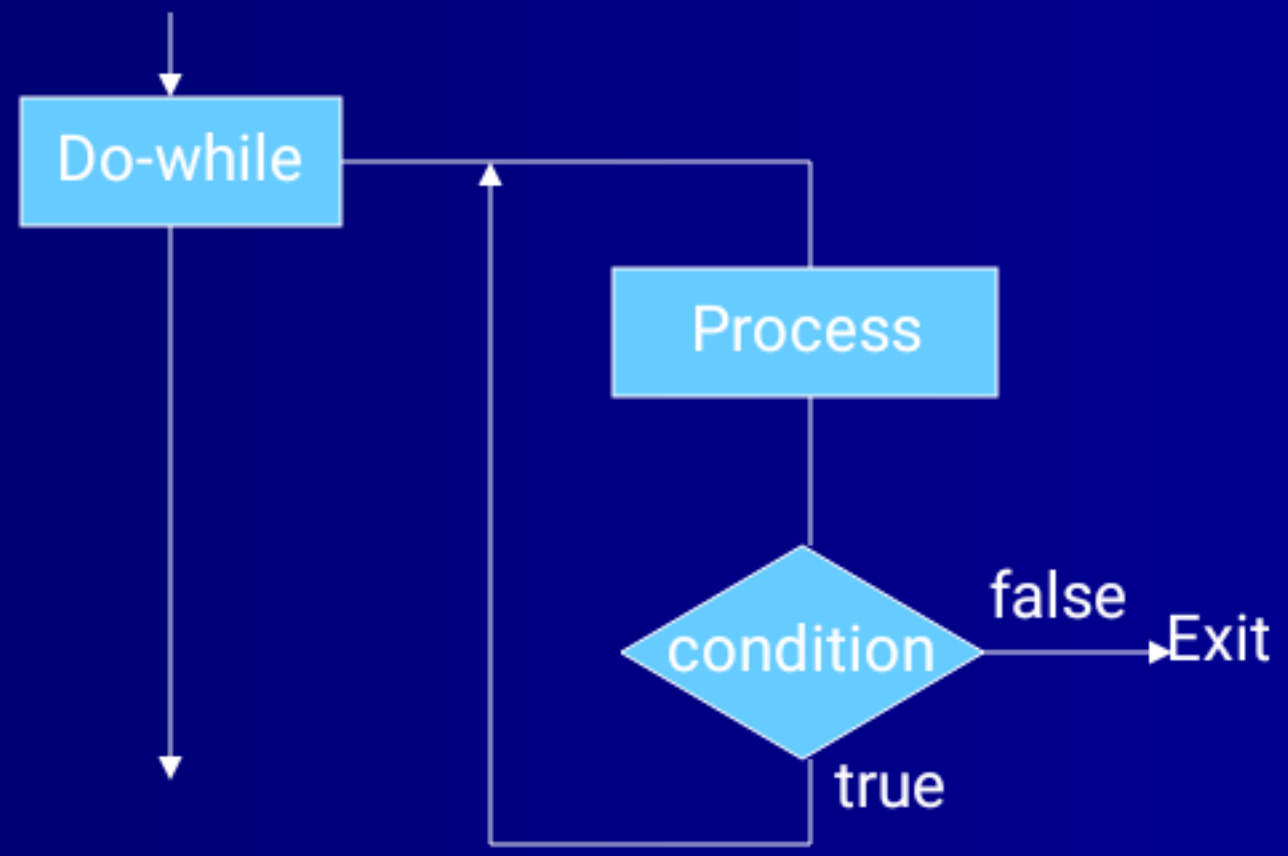
```
do
 {
   statements ;

 }
while ( condition ) ;
```

Example-Guessing game

```
char c ;
int tryNum = 1 ;
do
{
cout << "Please enter your guess by pressing a character key from a to z " ;
cin >> c ;
if ( c == 'z' )
  {
cout << "Congratulations! you guessed the right answer" ;
tryNum = 6 ;
}
else
tryNum  = tryNum + 1 ;
} while ( tryNum <= 5 ) ;
```

Flow chart for do-while loop



Relational Operators

```
char c ;
  int tryNum , maxTries ;
  tryNum = 1 ;
  maxTries = 5 ;
  cout << "Guess the alphabet between a to z " ;
  cin >> c ;
  while ( ( tryNum <= maxTries ) && ( c! = 'z' ) )
  {
   cout << "Guess the alphabet between a to z " ;
   cin >> c ;
   tryNum = tryNum + 1 ;
  }
```

for Loop

For loop

```
for ( initialization condition ; termination condition ; increment condition )
{
 statement ( s ) ;
}
```

Example

```
int counter ;
for( counter = 0 ; counter < 10 ; counter = counter + 1 )
        cout << counter;
```

Output

0123456789

Table for 2

2 x 1 = 2

2 x 2 = 4

2 x 3 = 6

⋮

⋮

2 x 10 = 20

Example - Calculate Table for 2

```
#include <iostream.h>
main ( )
{
 int counter ;
  for ( counter = 1 ; counter <= 10 ; counter = counter + 1 )
 {
      cout << "2 x " << counter << " = " << 2* counter << "\n" ;
    }
}
```

Output

2 x1 = 2

2 x 2 = 4

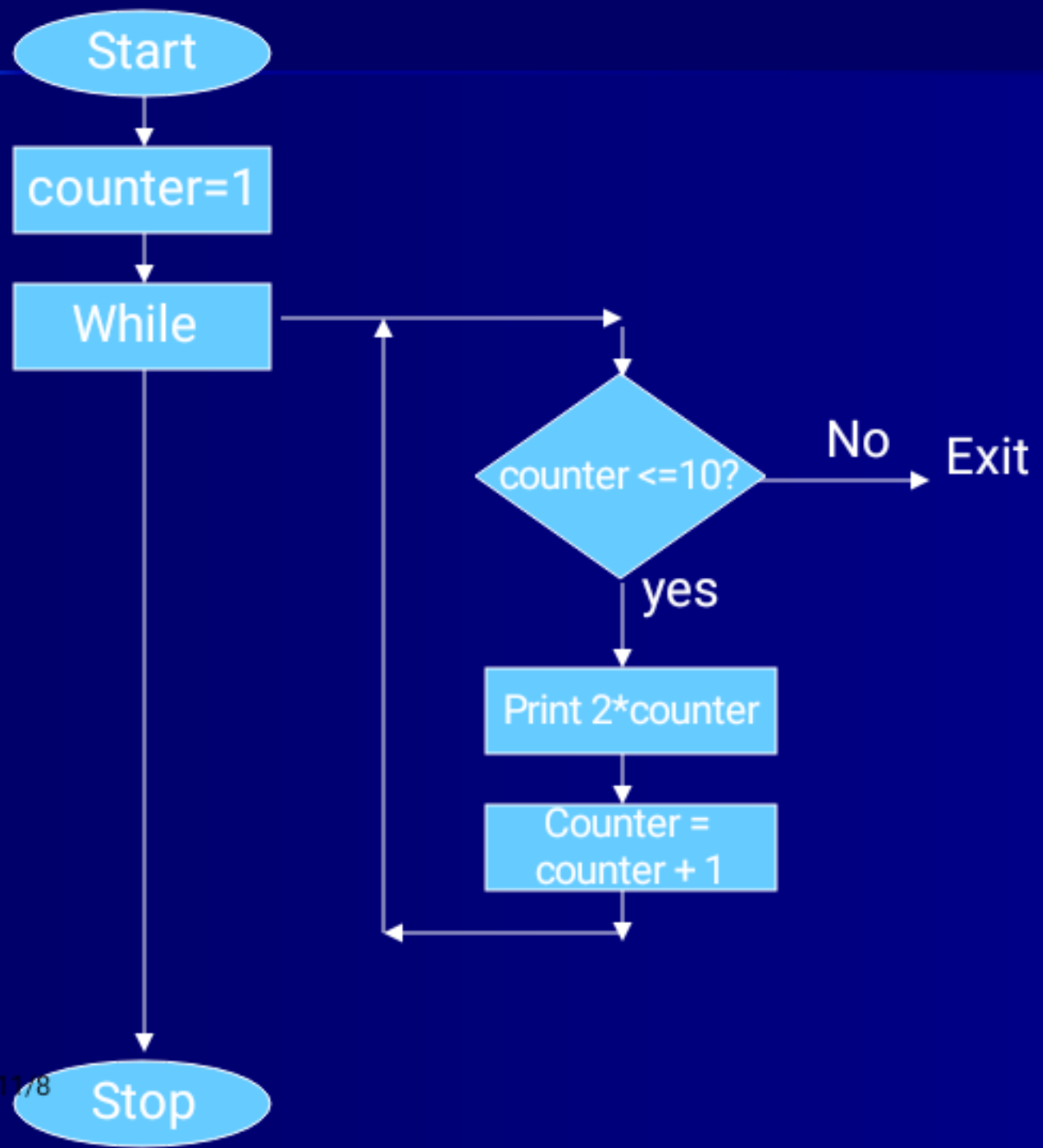
2 x 3 = 6

⋮

⋮

2 x 10 = 20

Flow chart for the 'Table' example



Example: Calculate Table- Enhanced

```
#include <iostream.h>
main ( )
{
 int number ;
  int maxMultiplier ;
  int counter ;
  maxMultiplier = 10 ;
  cout << " Please enter the number for which you wish to construct the table " ;
 cin >> number ;
 for ( counter = 1 ; counter <= maxMultiplier ; counter = counter + 1 )
 {
    cout << number <<" x " << counter<< " = " <<  number * counter << "\n" ;
 }
}
```

- Always think re-use
- Don’t use explicit constants

Increment operator

++

- counter ++ ;

  same as

- counter = counter + 1;

Decrement operator

--

- counter -- ;

  same as

- counter = counter - 1

+=

- counter += 3 ;

same as

- counter = counter + 3 ;

-=

- counter -= 5 ;

  same as

- counter = counter – 5 ;

*=

x*=2

x = x * 2

/=

x /= 2

x = x / 2

Compound Assignment Operators

operator=

%=

- x %= 2 ;

  same as

- x = x % 2 ;

Comments

- Write comment at the top program to show what it does
- Write comments that mean some thing

In today's lecture

- Do - while
  - Executes the code at least ones
- For loop
  - Executes at least zero times
- Short hand operators
  - Incrementing
  - Decrementing
- Compound assignment operator